LEARNER GUIDE

Composite shapes and design choices

Perimeter and Area · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **composite shape** is made of simpler shapes joined together, like an L-shaped room. To find its **area**, split it into rectangles and add their areas. To find its **perimeter**, add every outside edge. A missing length can be found from the edges on the opposite side, because together they match. Then you use area and perimeter to make design choices: how much flooring, how much skirting board, and which room is cheaper.

What you are learning to notice

- There is more than one way to split a shape, and every way gives the same total area.
- Sometimes taking away is quicker: the whole rectangle minus the missing corner.
- Walls on one side of a room add up to the wall opposite.
- Cutting a rectangular corner out of a rectangle changes its area but not its perimeter.
- 1 m = 100 cm, so 28 m of skirting board is 2800 cm.

Moving through it, one step at a time

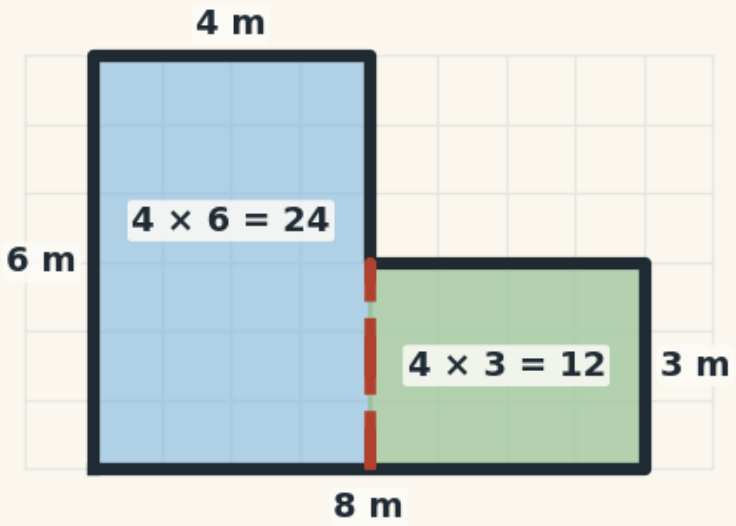
Area: draw a line that splits the shape into rectangles, find each area (length × width), and add. Perimeter: first find any missing walls from the walls opposite, then walk all the way round, adding every wall once. Check the area by splitting a different way.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Split an L-shaped room

An L-shaped room is 8 m along one wall and 6 m along another. Find its floor area and its perimeter.

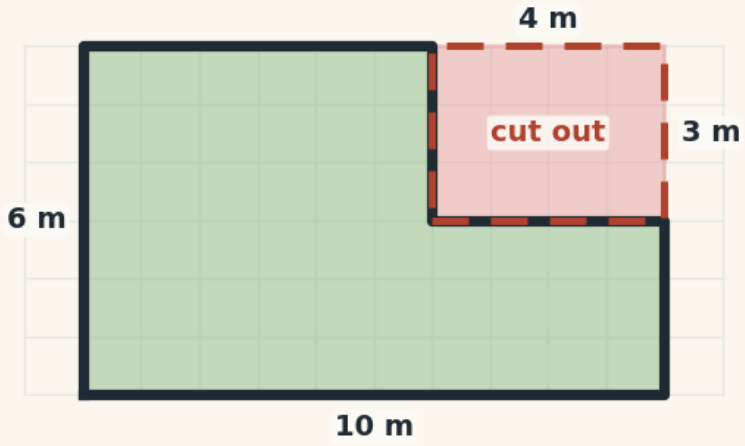


1. Cut down to make two rectangles.	$4 \times 6 = 24$ m^2 and $4 \times 3 = 12$ m^2
2. Add the pieces.	$24 + 12 = 36$ m^2
3. Missing walls: 8 – 4 and 6 – 3	4 m and 3 m
4. Walk round: 8 + 3 + 4 + 3 + 4 + 6	28 m

Floor 36 m^2. Walls 28 m (2800 cm of skirting board).

Example 2 — Take the missing corner away

A 10 m by 6 m room has a 4 m by 3 m corner cut out. Carpet costs £10 per m^2. What does the carpet cost?



1. The whole rectangle — $10 \times 6 = 60\ m^2$
2. The missing corner — $4 \times 3 = 12\ m^2$
3. The floor — $60 - 12 = 48\ m^2$
4. The carpet — $48 \times £10 = £480$

Floor 48 m^2; carpet £480.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (area of rectangles (Lesson 2) and metres and centimetres (Place Value Lesson 8)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

> Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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